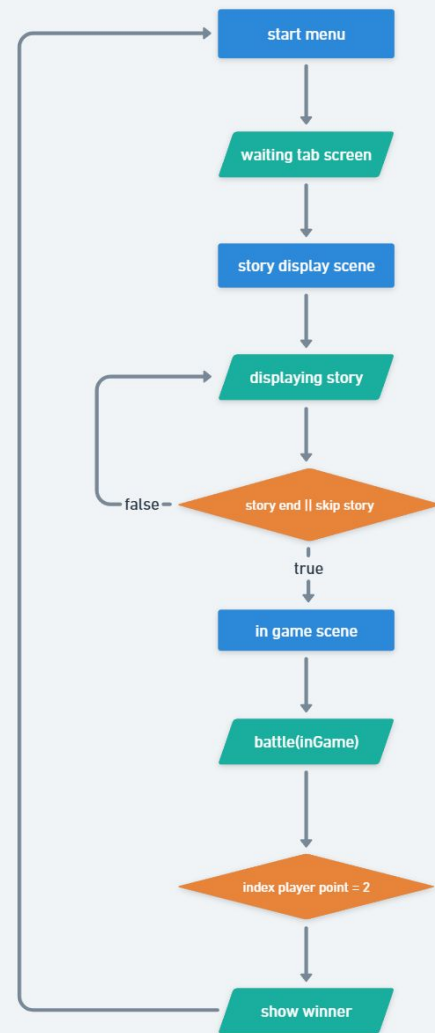


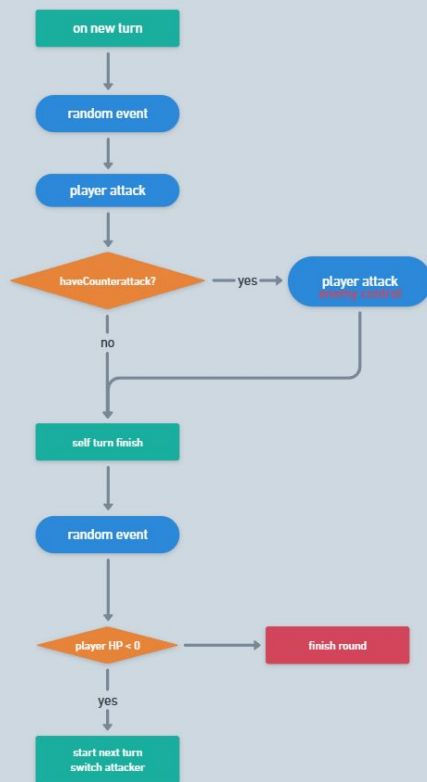
期中作業

111AC1013 范姜昱任

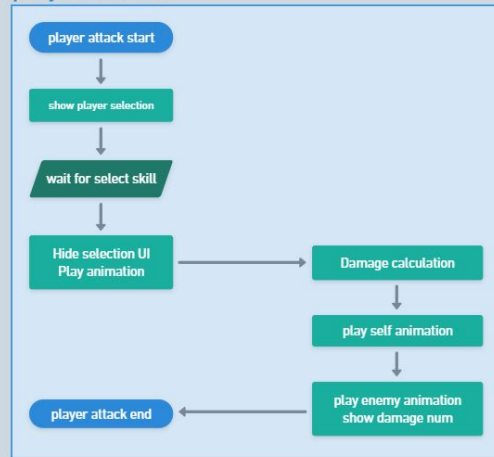
流程圖



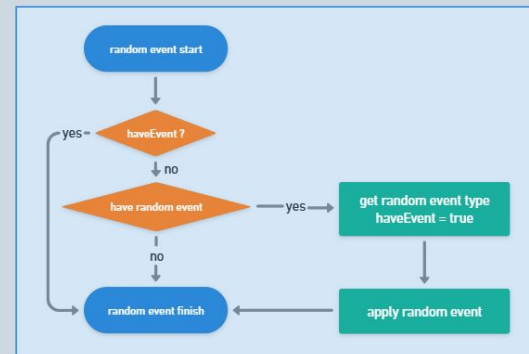
流程圖 (inGame)



player attack



random event



```

<MonoBehaviour>
inGame_UI_Managermanager

- gc:GameController
- p1:Player
- p2:Player
+ P1_Name:Text
+ P2_Name:Text
+ P1_HP:Text
+ P1_EP:Text
+ P2_HP:Text
+ P2_EP:Text
+ playerScore_UI:Text
+ roundCount_UI:Text
+ turnCount_UI:Text
+ hintUI_Animator:Animator
+ hintText:Text
+ P1_HpBar:Slider
+ P2_HpBar:Slider
+ P1_EpBar:Slider
+ P2_EpBar:Slider

```

```

- updateUI():void
- HpBarUpdate():void
- HpBarUpdate():void
+ displayGameStart():async Task
+ displayRoundStart(int):async Task
+ displayNowTurnCount(int,bool):async Task
+ displayNowAttackerDefender(string,string):async Task
+ displayCounterAttacker(string):async Task
+ displayRoundOver(int):async Task
+ displayWinner(string):async Task
+ displayOnEvent(string):async Task
+ EventApplyAnimation(Player,EeventType):async Task
+ displayAddActorHP(Player,int):async Task
+ displayDecreaseActorHP(Player,int):async Task
+ displayAddActorEP(Player,int):async Task
+ displayDecreaseActorEP(Player,int):async Task
+ displayAddActorATK(Player,int):async Task
+ displayAddActorActorDEF(Player,int):async Task
- Update():void

```

```

<MonoBehaviour>
UI_manager

+ canvases:Canvas[]
- initialPositions: Vector3[]
+ centerObject:Transform
+ nowCanvasName:string
+ transitionAnimator:Animator

+ SwitchCanvas(string):void
- PlayTransitionAndSwitchCanvas(string):IEnumerator
- MoveCanvasToCenter(Canvas):void
- ResetCanvasPositions():void
- ExitGame():void

```

```

<MonoBehaviour>
attackEventAnimation

+ attacker:Player
+ defender:Player
+ skill:ActorSkill
+ cam:gameCamera
+ gameUI:inGameUI_Manager
+ damage:int
+ isCounterAttack:bool

+ playAnimation():async Task

```

```

<MonoBehaviour>
gameCamera

+ cam:Camera
+ animator:Animator
+ target :Vector2
+ db:debugUI_manager

+ zoomInToPlayer():async Task
+ cameraPanToPlayer():async Task
+ cameraReplacel():async Task

```

```

<MonoBehaviour>
GameController

+ ul_Manager:UI_manager
+ gCamera:gameCamera
+ gameUI:inGameUI_Manager
- endGame:bool
- endRound:bool
+ players: Player[2]
- player1:Player
- player2:Player
+ skillButtons:Button[3]
+ restButton:Button
+ skillPanel:GameObject
+ skillsActorName:Text
+ currentPlayer:Player
- opponent:Player
+ isGaming:bool
+ round :int
+ turnCount :int
+ secondHalf:bool
+ alreadyCounterAttack:bool
+ buttonSelect:bool
- skillChoice:int

+ onClickStartGame():async void
+ StartGame():async Task
- StartRound():async Task
- StartTurn():async Task
- waitPlayerSelect():async Task
- OnButtonClicked(int):void
- AttackAction(int):async Task
- TriggerRandomEvent():async Task
- showActorSkillOption(Player):void
- PlayerAttack(Player,Player,int,bool):async Task
+ CalculateDamage(int,int,int):int
- endDetect():async Task
- endRoundDetect():bool
- EndRound():async Task
- EndGame():async Task
- AlivePlayer():string
- addScort(string):void
- Update():void

```

```

<MonoBehaviour>
debugUI_manager

+ debugUI:GameObject
- debugUIActive:bool
+ gc:GameController
+ UI_Manager:UI_manager
+ P1_data:Text
+ P2_data:Text
+ gameLog:Text
+ actorLog:Text
+ camLog:Text
+ nowCanvas:Text
+ RoundTurn:Text

```

```

+ updateDebugUI():void
+ GameLog():void
+ ActorLog():void
+ CamLog():void
+ gamelnitial():void
- Update():void

```

```

<MonoBehaviour>
Player

+ loader:jsonLoader
+ animator:Animator
+ useActorName:string
+ actorIMG:SpriteRenderer
+ actor:Actor
+ score:int
+ isTurn:bool
+ isEvent:bool
+ playerSelect:Dropdown
- playerPos:Vector2

```

```

+ selectActor():void
+ onTurn():void
+ endTurn():void
+ roundInitial():void
+ gamelnitial():void
+ ChangeColorByActorName():void
+ cameraOnEnemy(Player):void
+ moveToEnemy(Player):void
+ moveBack():void
+ costEP(int):void
+ ApplyDamage(int):void
+ getRandomEventType():EeventType
+ ApplyRandomType(EeventType):void

```

```

class ActorSkill

+ name:string
+ atk:int
+ epCost:int
+ hpRecover:int
+ epRecover:int
+ deflIncrease:int
+ canBeCountered:bool

```

```

class SkillData

+ skills:ActorSkill[]

```

```

jsonLoader

+ actors:List<Actor>
+ skills:List<ActorSkill>
+ eevents:List<EeventType>

+ LoadSkills():void
+ LoadActors():void
+ LoadEvent():void

```

```

class Actor

+ name:string
+ hp:int
+ ep:int
+ atk:int
+ def:int
+ actorSkill_1:string
+ actorSkill_2:string
+ actorSkill_3:string
+ skills:ActorSkill[]

```

```

class actorData

+ actors:List<Actor>

```

```

class EventType

+ name:string
+ discribs:string
+ increaseHp:int
+ increaseEp:int
+ increaseATK:int
+ increaseDEF:int

```

```

class EventData

+ Eevents:List<EeventType>

```

遊戲機制：

四種角色(四選二)

六種技能(每隻角色三個)

七種隨機事件(觸發率40%)

先獲得2分者獲勝

近戰攻擊會觸發反擊，遠程攻擊傷害較低

EP不夠時可以休息回復體力(動畫最短)

[程式碼連結](https://drive.google.com/drive/folders/12IkSRb30mB1CvIlyLqgJWWWhQkSV0IF4w?usp=sharing)

<https://drive.google.com/drive/folders/12IkSRb30mB1CvIlyLqgJWWWhQkSV0IF4w?usp=sharing>

